Macro-Driven Credit Portfolio Stress Testing with Live Economic Data

Overview

This project extends a Credit Probability of Default (PD) model by simulating how portfolio credit risk behaves under macroeconomic stress scenarios using live economic data. It implements the Basel II/III Vasicek single-factor model for stress testing and provides comprehensive risk analysis, visualization, and reporting.

Key Features

- Live Macro Data: Automatically fetches current unemployment, GDP, inflation, and interest rate data from FRED
- Stress Scenarios: Creates Baseline, Adverse, and Severely Adverse scenarios based on current economic conditions
- Vasicek Model: Uses the Basel II/III single-factor model for stress testing
- Risk Migration: Tracks how borrowers move across risk bands under stress
- Capital Impact: Estimates Basel IRB risk-weighted assets (RWA) impact
- No API Keys Required: Uses pandas_datareader to fetch public FRED data
- Comprehensive Dashboard: Interactive visualizations for all key metrics
- Automated Reporting: Summary report with actionable recommendations

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Credit_Risk_Stress_Testing/
|
├── README.md
├── requirements.txt
├── environment.yml
|
├── notebooks/
|   └── Macro_Driven_Stress_Testing.ipynb
|
├── data/
|   └── credit_risk_csv.csv
|
├── outputs/
```

```
|   ├── stress_test_results_live_data.csv
|   ├── default_rate_home_ownership.png
|   ├── default_rate_loan_grade.png
|   ├── lti_ratio_analysis.png
|   └── stress_test_dashboard.html
|
├── src/
|   ├── data_loader.py
|   ├── feature_engineering.py
|   ├── stress_testing.py
|   └── visualization.py
|
└── docs/
    ├── project_overview.md
    └── api_reference.md
```

Key Results

Stress Test Summary

Metric	Baseline	Adverse	Severely Adverse
Average PD	22%	25%	34%
Expected Loss	$34.6M	$39.0M	$51.8M
EL Increase	-	+13%	+51%
High-Risk Borrowers	6,200	7,100	10,400
Risk-Weighted Assets	$153.8M	$222.4M	$366.7M
RWA Increase	-	+45%	+138%

Risk Band Migration

Risk Band	Baseline	Adverse	Severely Adverse
Very Low Risk	15,284	8,972	1,872
Low Risk	8,417	11,802	5,078
Moderate Risk	2,687	4,732	15,215
High Risk	778	1,407	3,877
Very High Risk	5,415	5,668	6,539

Dependencies

Core Dependencies

- Python >= 3.8
- numpy >= 1.21.0
- pandas >= 1.3.0
- matplotlib >= 3.4.0
- seaborn >= 0.11.0
- scikit-learn >= 1.0.0
- scipy >= 1.7.0
- xgboost >= 1.5.0
- shap >= 0.41.0
- pandas-datareader >= 0.10.0

Optional Dependencies

- plotly >= 5.0.0 (interactive visualizations)
- ipywidgets >= 7.6.0 (interactive widgets)
- jupyter >= 1.0.0 (Jupyter notebook)

Mathematical Models

Vasicek Single-Factor Model

PD_stressed = PHI((PHI_inv(PD_baseline) - sqrt(rho) * Z) / sqrt(1-rho))

Where:

- PD_baseline: Baseline Probability of Default
- PD_stressed: Stressed Probability of Default
- rho: Asset Correlation (0.03 to 0.16)
- Z: Systematic Macro Factor
- PHI: Standard Normal CDF

Expected Loss

EL = PD * LGD * EAD

Where:

- LGD: Loss Given Default (assumed 45%)
- EAD: Exposure at Default (loan_amnt)
- EL: Expected Loss

Basel IRB Risk Weight

K = LGD * PHI((PHI_inv(PD) + sqrt(rho) * PHI_inv(0.999)) / sqrt(1-rho)) - LGD * PD
RW = K * 12.5 * MA

Where:

- K: Capital Requirement
- RW: Risk Weight
- MA: Maturity Adjustment (assumed 1.0)
- PHI_inv(0.999): 99.9% confidence level